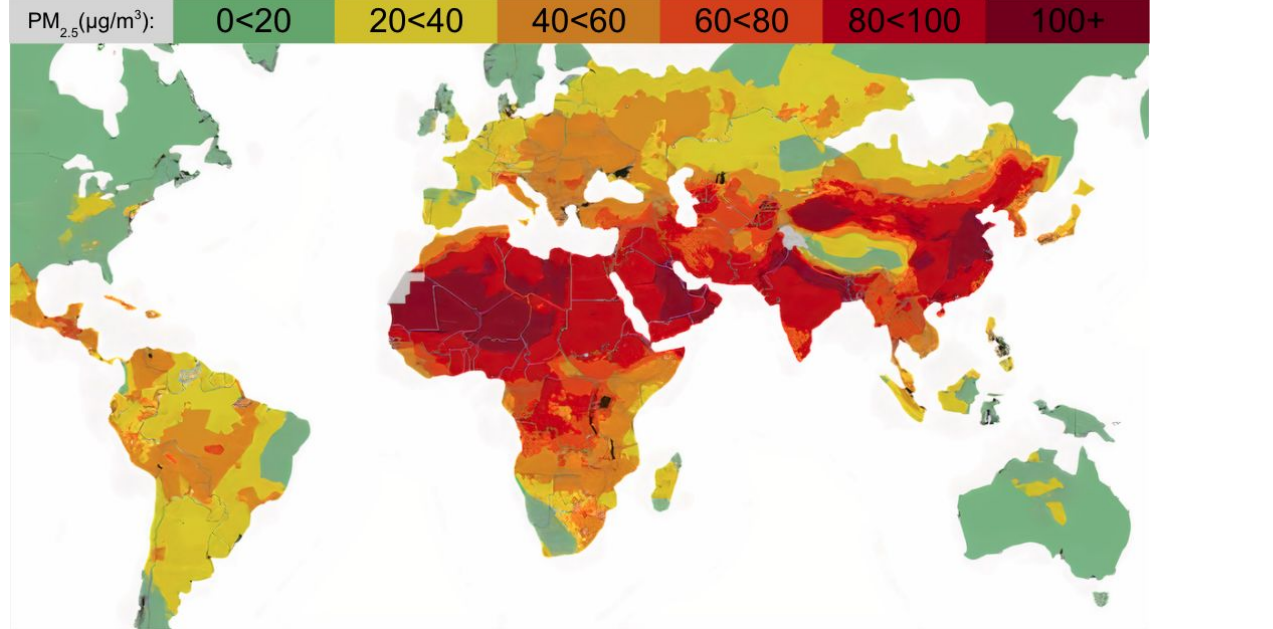


Introduction and Hypothesis

There are few things more important to your health than the air that sustains your body with each breath, but as the air quality crisis continues to impose its effects around the world, the population of people finding themselves without access to unpolluted air is increasing drastically. The solution to this lies in the early detection and feedback of polluted air, especially when people are unaware of the quality of the air they are breathing. The Worldwide Health Organization estimates that 90% of people worldwide breathe polluted air and and causes over 6 million deaths in developing nations yearly. For a problem that significantly impacts so many people across the globe, it is shocking that there are no current technological solutions that give individuals and households the ability to accurately measure and understand the quality of the air they breathe in.

Currently, the widely used Air Quality Index (AQI) is used to calculate a region's air quality with one to four air pollutants. However, there are far more than 4 critical air pollutants and it is understood that air quality can change gradually from environmental factors or rapidly from manmade factors in small scale locations. Despite its importance, air quality measurement has received minimal technological innovation as current monitors lack accuracy and measurable metrics and can cost thousands of dollars. It is clearly evident that there is significant room for improvement in air quality analysis, both in its accuracy and accessibility. For a fraction of the cost, single compound sensors can be enhanced with artificial intelligence analysis to accurately measure, analyze, and predict changes in the air around us more effectively and efficiently.

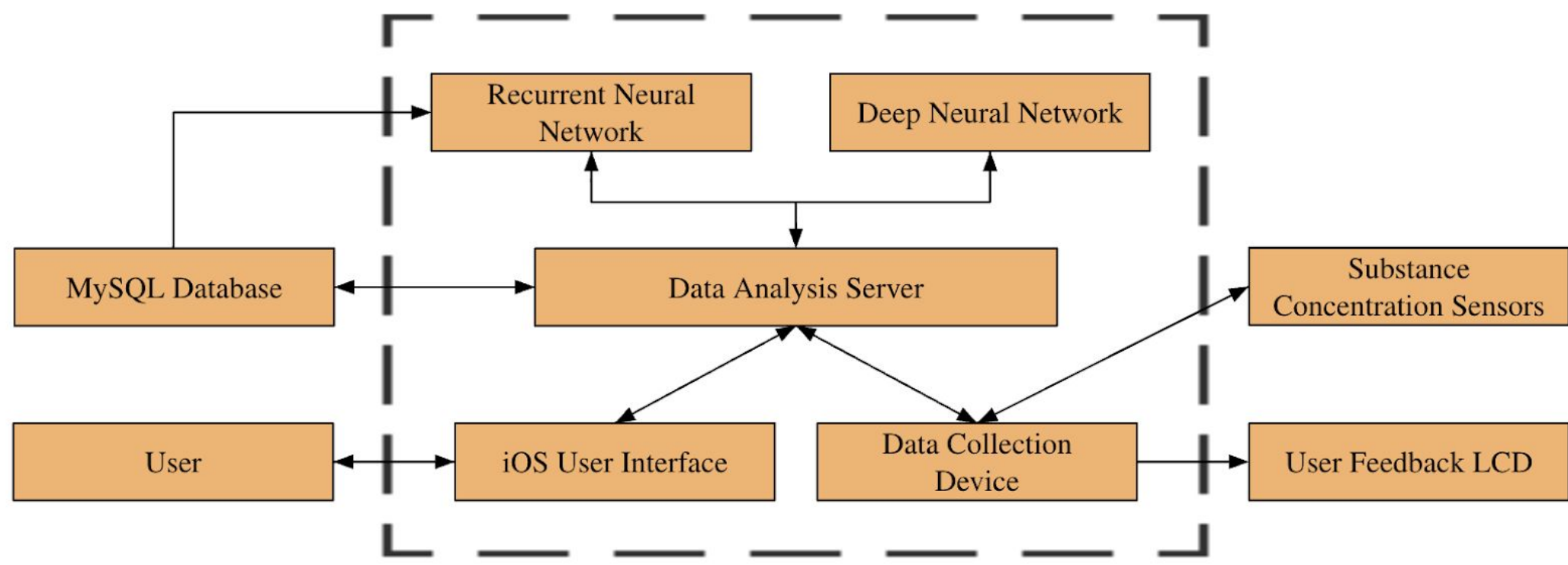


Alternate Hypothesis (H₁): The constructed system will accurately make predictive calculations of air quality changes from environmental substance concentrations.

Engineering Goals

	Maximize Prediction Accuracy	The measurement system will calculate a predictive aggregate air quality score that approaches an accurate representation of reality.
	Accessible Metrics and Trends	Trends in air quality predictions and collected environmental data points will be easily accessible and accurately interpreted.
	Affordable for Household Consumer	The constructed system will be designed for maximum affordability and household utilization.

System Architecture



The constructed air quality assessment system consists of four independent components: Data Collection Device, Data Analysis Server, Recurrent and Deep Neural Network, and iOS User Interface. All data transfer and communication is made through a user's Local Area Network (LAN) to connect through a central router to the specified internet protocol (IP) address of the Data Analysis Server. Similarly, in this system, all requests of the iOS User Interface for substance concentrations, graphs, database entries, and RNN/DNN analysis are made via parameter-containing representational state transfer (REST) application programming interface (API) request to an authenticated MySQL Database and sent over the internet allowing for faster and higher-capacity data communication.

Merge Model Predictions

Because the developed device utilizes 2 neural networks, it requires an algorithm to merge the predictions of both models into a final prediction, which represents the aggregate air quality score (AAQS). It must also quantify the resulting loss, which measures the difference between a prediction and reality. To accomplish this, the losses of the individual models are used to calculate a weight for each model that represents its significance to the final prediction. Because a lower loss represents a more accurate prediction, the algorithm values a prediction with a lower margin of error more than a prediction with a higher margin of error accordingly.

Weight of Model Significance

$$W_X = L_X * \frac{L_Y}{L_D + L_R}$$

Calculate Final Prediction

$$P_{final} = P_R * W_R + P_D * P_D$$

Bounds of Prediction Loss

$$[A_X, Z_X] = [P_X * (1 - L_X), P_X * (1 + L_X)]$$

Calculate Final Loss

$$L_{max} = MAX(\frac{P_{final}}{A_R * W_R + A_D * W_D}, \frac{P_{final}}{Z_R * W_R + Z_D * W_D})$$

The weight of each model (W_x) is calculated to quantify the significance of the model's metrics by using the loss of the corresponding model (L_x) and the other neural network model (L_y).

The compound neural network's final prediction (P_{final}) representing the AAQS can be calculated by taking a weighted average of the model prediction (P_R, P_D) with its corresponding weight (W_R, W_D), which adjusts based on each model's performance.

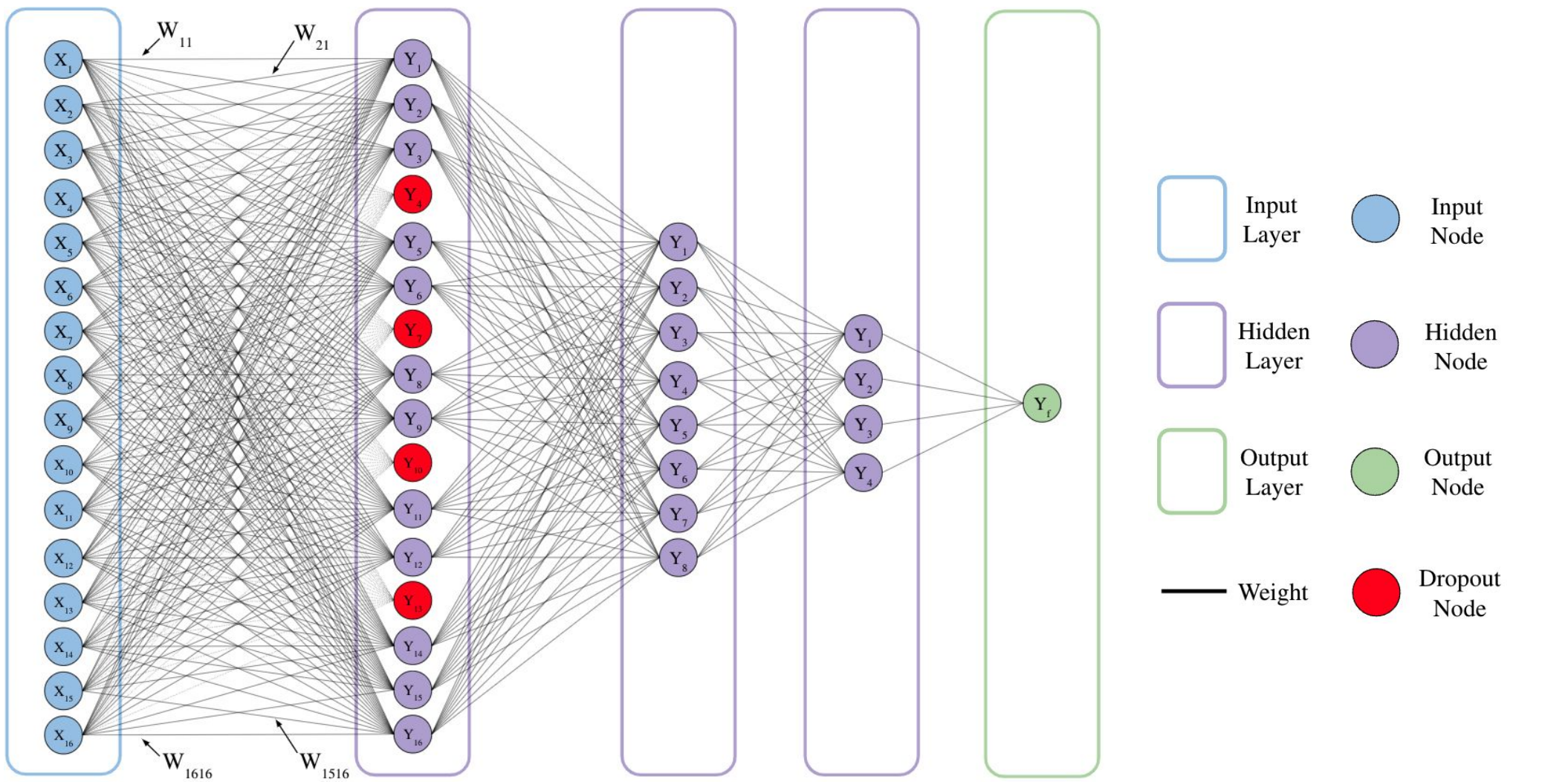
Quantify the lower bound (A_x) and upper bound (Z_x) of the AAQS for each model. To do this, each model prediction (P_x) is scaled by its loss (L_x) in both directions.

The algorithm calculates the max loss in the final compound prediction if each prediction is underestimated or overestimated, and the greater value is considered the maximum loss.

Early Detection and Insight into Air Quality Degradation Using a Compound Neural Network-Based Substance Concentration Measurement System

Compound Machine Learning Approach

Deep Neural Network (DNN)



A deep neural network (DNN) is constructed to best determine the correlation between individual substance concentrations. It was programmed in Python using the Google Tensorflow Machine Learning library. The DNN operates by passing substance concentration data through multiple hidden layers which are focused on isolating intervariable trends between parameters. It outputs a final value which is representative of the predicted air quality score. The DNN is trained on the server launch and a prediction iteration is made every time a new data point is received. As the data set grows, the DNN can become increasingly accurate in its predictions.

Hidden Layer 1 - Feed Forward

$$Y_n = W^T X$$

First, the information is passed to the next node. This is done by multiplying each node's value (X) by each of the weight (rows of W^T) and sending it to every node (Y_n) in the next layer. This ensures that every combination of node value and weight is created.

Hidden Layer 2 - Backpropagation

$$E_{n-1} = W_n^T \cdot E_n$$

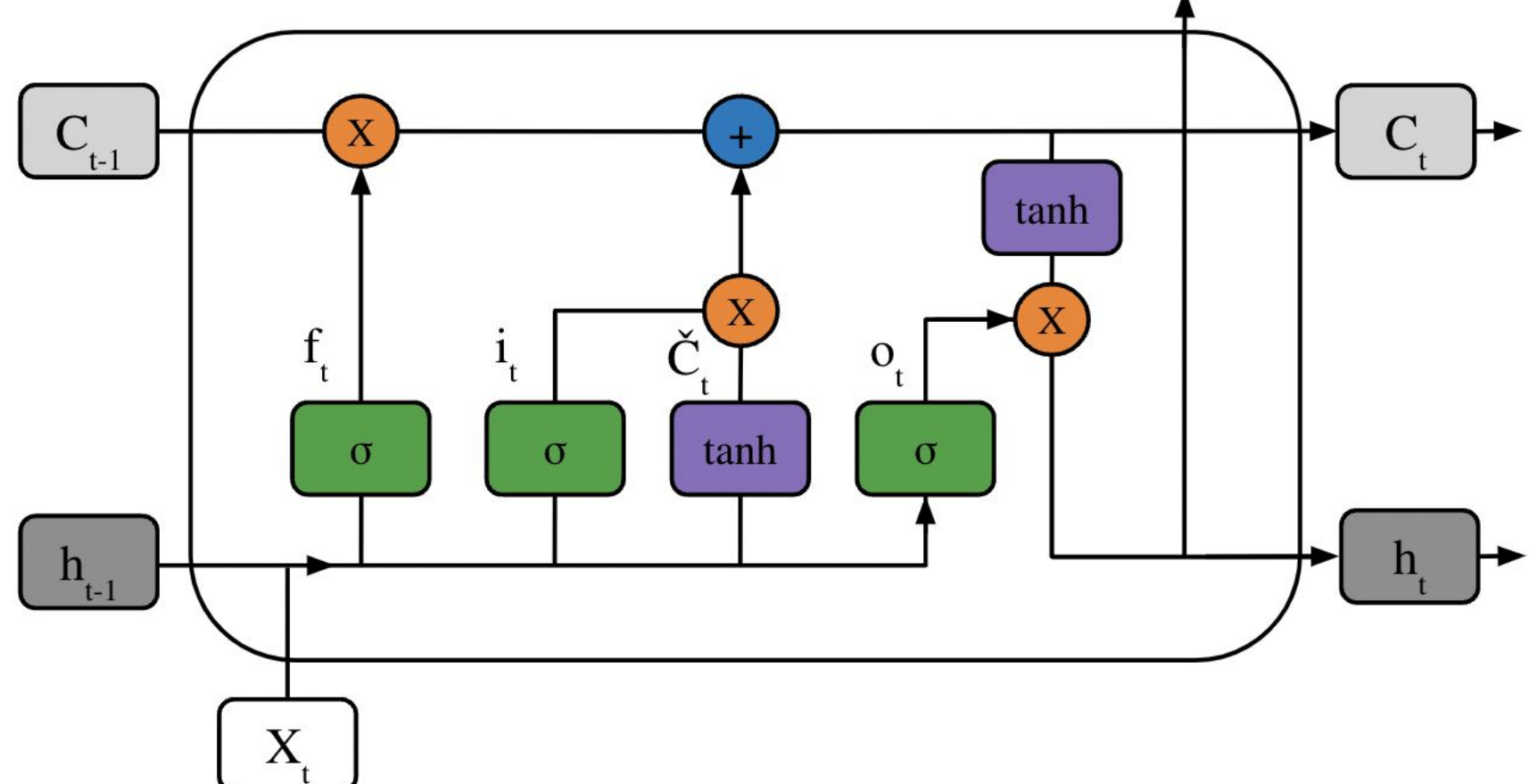
In the second hidden layer, the errors of the weights are calculated and sent back to the previous layer's nodes. The error of each input node (E_{n-1}) is equivalent to the product of a transposed weight matrix (W_n^T) and the error of the output node (E_n).

Hidden Layer 3 - Update Weights

$$W_n = W_n - Lr \cdot \frac{\delta E_{n+1}}{\delta W_n}$$

Finally, the weights are updated before being sent to the next layer. The weights are updated (W_n) according to the partial derivative of the error function with respect to weight (δE_{n+1}/δW_n) and each nodes error (n+1), which is scaled by the model's learning rate (Lr).

Recurrent Neural Network (RNN)



The recurrent neural network (RNN) is programmed in Python using the Google Tensorflow Machine Learning library. The RNN accepts timestamps as an additional parameter and factors chronological progression into predictions, which is important for early detection of poor air quality. The RNN is trained on server deployment, and a new prediction iteration is run every time new data is received. In each layer of the RNN, various mathematical formulas are used to output the final prediction.

Hidden Layer 1 - Forget Gate

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

First, a sigmoid function uses the node's weight coefficient (W_f), outputs (h_{t-1}), new inputs (x_t), and a defined constant (b_f) to generate a value in [0, 1] representative of the percentage of the data to be kept and passed into the next layer.

Hidden Layer 2 - Cell State Storage

$$i_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

Next, a sigmoid function determines the amount that values will be updated in the next layer. After this, a hyperbolic tangent function is used to create a vector of new candidate values for the cell state (C̃_t).

Hidden Layer 3 - Cell State Update

$$C_t = (f_t \cdot C_{t-1}) + (i_t \cdot \tilde{C}_t)$$

At each time step in the data, previous function outputs are used to calculate the new values of the cell state by multiplying the number of the kept values (f_t) and old cell state (C_{t-1}), and adding it to the product of the updated values (i_t) and candidate values (C̃_t).

Hidden Layer 4 - Calculate Output

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

A sigmoid function is used to calculate proportion of the final output to be kept. Then, the current cell state is entered into a hyperbolic tangent function multiplied by the output from the previous step (o_t) to get the final output of the RNN LSTM cell (h_t).

Hardware Construction & Analysis

Functionality

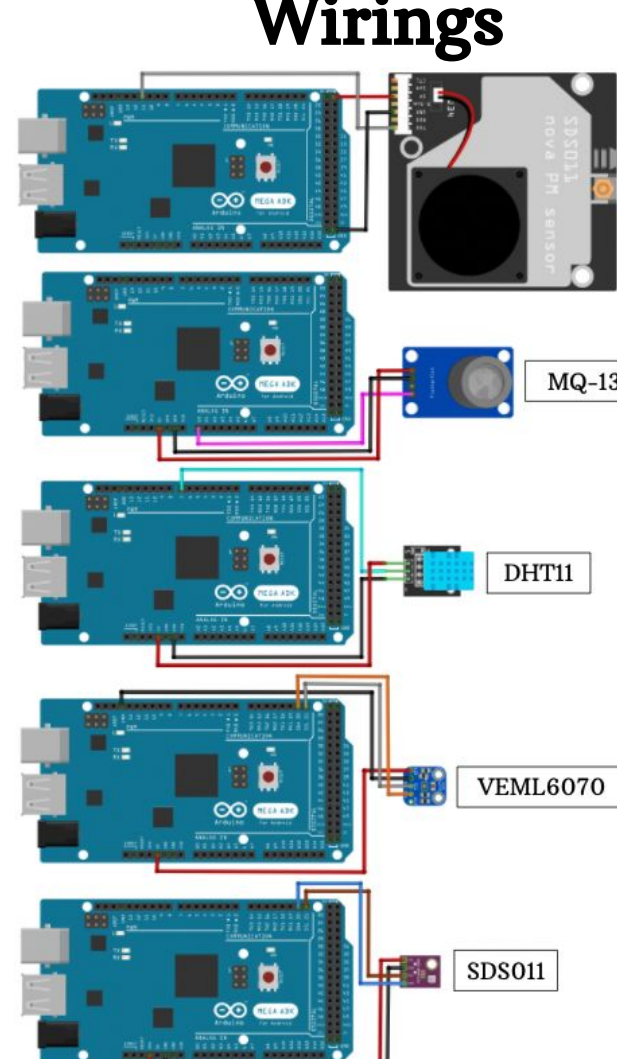
The Arduino-based data collection device (DCD) is programmed in Arduino C and constructed to measure the concentrations of seven individual substances in the air. The device establishes a connection to a WIFI network on start. Every 60 seconds, these concentrations are sent via HTTP requests to an analysis server, which utilizes machine learning to continuously predict the air quality at several points in the future.

The user has access to the raw data collected by the DCD by accessing the Swift application on their mobile devices. The liquid crystal display on the DCD also gives feedback of the returned AAQS.

Sensor Inputs

Sensor	Value Range	Measures
SDS011	0-999.9 µm/m³	Particulate Matter 2.5 & 10
MQ-135	0 - 300 ppm	Nitrogen Dioxide
DHT11	20% - 90% 0°C - 50°C	Temperature & Humidity
VEML6070	0 - 500	Ultraviolet Concentration
BME280	300 - 1100 hPa	Air Pressure

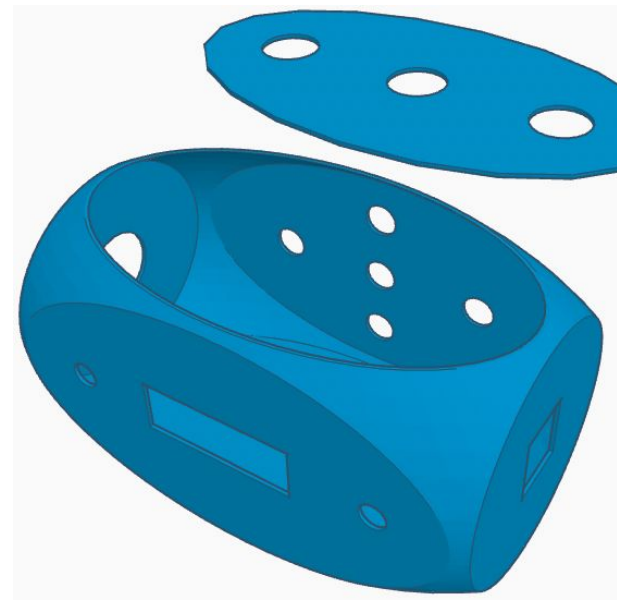
Individual Sensor Wirings



Combine into single circuit and provide common 5V and GND rails

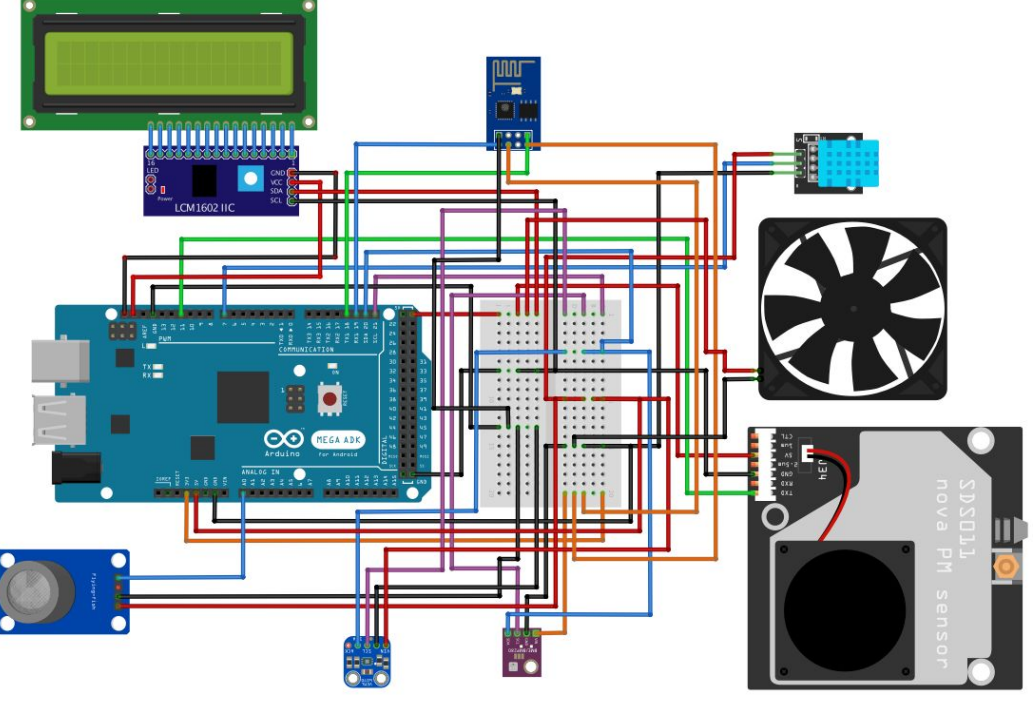
Enclose circuit in 3D-printed casing

3D-Modeled Case

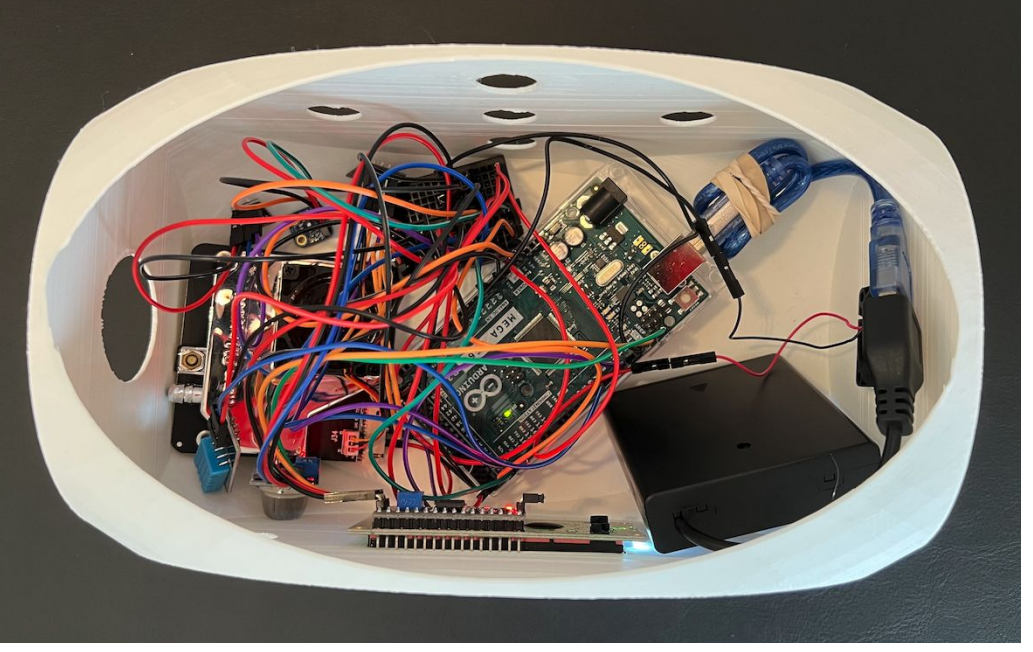


Final prototype contains all hardware components and enables air flow without exposing live electrical parts to user

Wiring Diagram of Complete Arduino Based Circuit

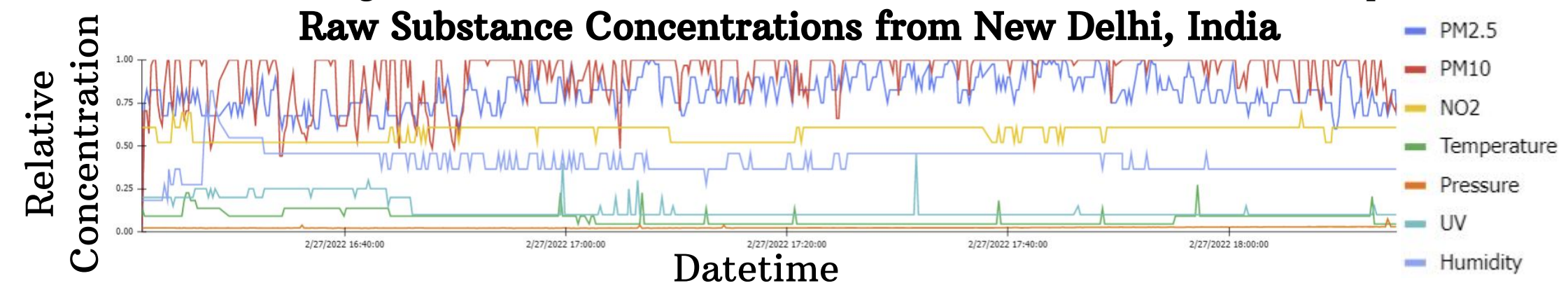


Constructed Data Collection Device (DCD)



Empirical Data from New Delhi

The data collection device was used to collect a raw data set of 7 individual substance concentrations from New Delhi, India. It is apparent that many air quality components fluctuate in concentration at a rapid rate over the course of a single day. Turning this data into an XY graph with overlapped substances allowed for the variation in the data to be better visualized and recognized. The y-axis represents 7 unique substance concentrations as a function of time scaled between the relative minimum mapped to 0 and relative maximum mapped to 1. This enables a visualization of the magnitude of fluctuation of each substance over the measured time period.

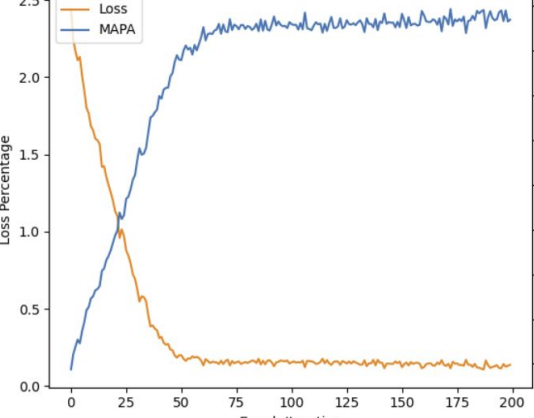


Neural Network Testing

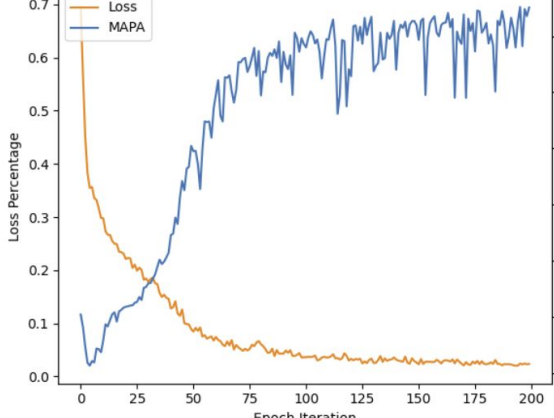
Training Data

To train the DNN and RNN, public air quality data from the United States Environmental Protection Agency (USEPA) and the World Air Quality Project (WAQ) was downloaded. To train the DNN, four substance concentration datasets were downloaded from the USEPA which contains the average substance concentrations for highly populated regions in America (N=336) over four years. For the RNN, chronological air quality data of New York City, Tokyo, New Delhi and Lagos (N=2482) was utilized. Success, loss and accuracy metrics are used as the primary model metrics.

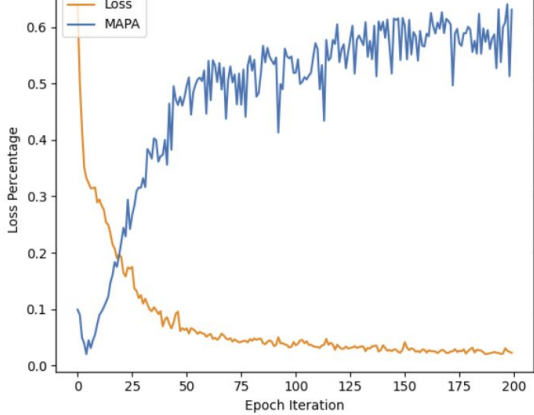
USEPA 2018 Model



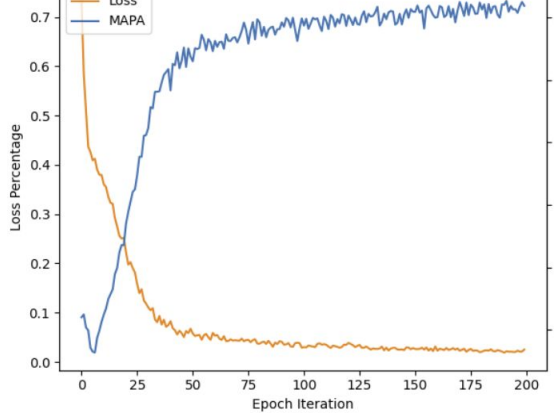
USEPA 2019 Model



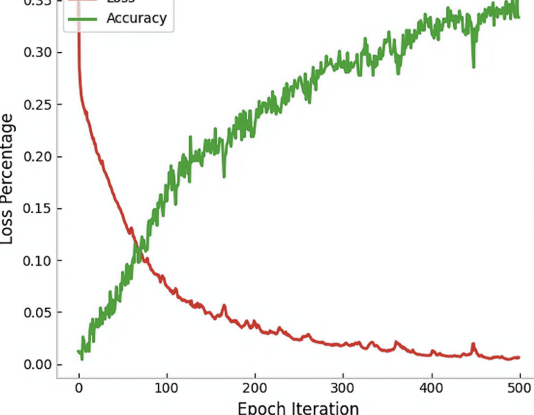
USEPA 2020 Model



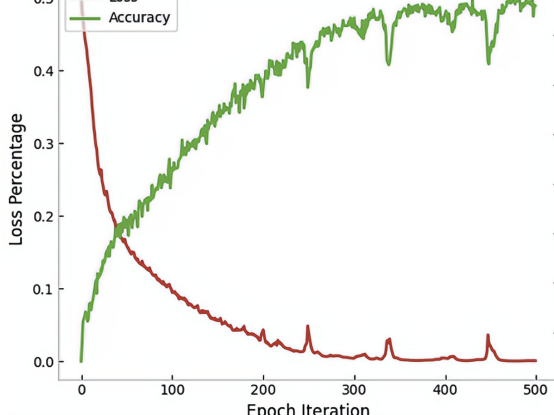
USEPA 2021 Model



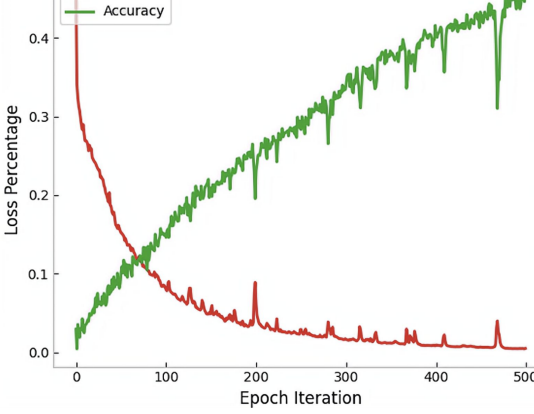
New York City Model



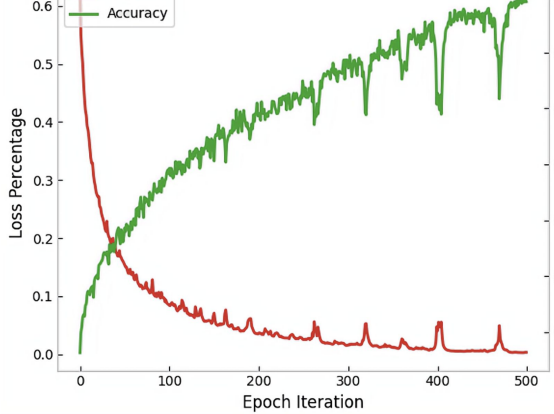
Tokyo, Japan Model



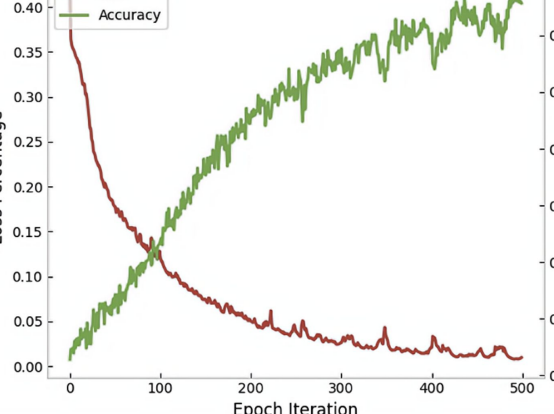
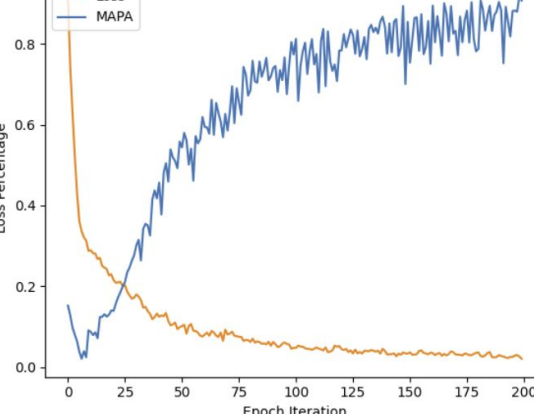
New Delhi, India Model



Lagos, Nigeria Model



Testing Data



Raw substance concentration data collected using the DCD in New Delhi, India was used to verify the ability of the DNN and RNN to accurately predict real-world AAQI figures. The models proved to be capable of making accurate predictions in a major city with rapidly fluctuating air quality.

Compound Neural Network Test

The compound neural network scales and merges predictions from the DNN and RNN into a final aggregate air quality score prediction. The epochs of each model were regularized over sub-sections, which are considered a compound model epochs. After training the model, the maximum loss was calculated to 1.27% and accuracy was calculated to 91.13%. Its progression was also visualized at each subsection to verify the effectiveness of compound neural network in making accurate predictions.

Findings and Implications

Raw data collected using the constructed DCD was paired with public air quality data to create training data sets for the DNN and RNN. Testing of these models yielded an overall accuracy of 91.13% and a loss of 1.27%. The Alternate Hypothesis (H₁) was supported by all of the data.

The implications of these findings are very significant. Although an extremely large portion of the world population is affected by poor air quality, most people do not have access to specific, low-level air quality data that truly describes the quality air they are breathing in. Although, the AQI is available for broad regions, it is not specific or accurate, and air quality can vary much more on a smaller scale. However, the accuracy of the compound neural network renders the system capable of accurately predicting air quality changes.

This affordable system can be readily implemented into any indoor or outdoor place, similar to the use of thermostats, smoke detectors, and carbon monoxide detectors, to help individuals take proactive measures to prevent inhaling polluted air. As current air quality monitoring devices, which are still inaccurate and imprecise, can cost over \$2,000, this For a widespread problem this system can provide a solution to people around the world regardless of wealth.

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